

**GRAPH THEORY:
HOMEWORK 3
BOOK SECTION 1.3.**

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1. Determine the maximum number of edges in a bipartite subgraph of P_n , C_n , and K_n . (Hint: for K_n it may be helpful to review the exercise from §1.1 on counting edges of bipartite graphs.)

Solution. Since P_n is bipartite, all of its $n - 1$ edges may be retained. Thus the maximum is $n - 1$.

If n is even, then C_n is bipartite, so the maximum is n . If n is odd, then C_n is not bipartite, so at least one edge must be omitted. Deleting any one edge leaves a path with $n - 1$ edges, which is bipartite. Hence the maximum is $n - 1$ when n is odd.

Finally, let a bipartite subgraph of K_n have partite sets of sizes a and b . We may include isolated vertices in the two sets, so we can assume $a + b = n$. Every edge goes between the two sets, and hence there are at most

$$ab \leq \left\lfloor \frac{n^2}{4} \right\rfloor$$

edges; the product is largest when the two integers are as equal as possible. This bound is attained by the subgraph $K_{\lfloor n/2 \rfloor, \lfloor n/2 \rfloor}$. Therefore the desired maximum for K_n is $\lfloor n^2/4 \rfloor$. $\square$

2. Prove that an even graph has no cut-edge

Proof. Suppose, to the contrary, that an even graph G has a cut-edge $e = uv$. In $G - e$, let H be the component containing u . Every vertex of H other than u has the same (even) degree in H as in G , while u has degree $d_G(u) - 1$, which is odd. Thus H has exactly one vertex of odd degree. This contradicts the Handshaking Lemma, according to which every graph has an even number of odd-degree vertices. Hence an even graph has no cut-edge. $\square$

3. The problem involves the hypercube Q_k .

- (a) Use the recursive description of Q_k (Example 1.3.8) to prove that $|e(Q_k)| = k2^{k-1}$
- (b) Prove that $K_2, 3$ is not contained in any hypercube Q_k . (Hint: prove that two distinct vertices in Q_k cannot have three common neighbors.)

Proof. (a) The recursive construction of Q_k takes two copies of Q_{k-1} and adds a matching joining corresponding vertices. Since Q_{k-1} has 2^{k-1} vertices, this gives

$$|E(Q_k)| = 2|E(Q_{k-1})| + 2^{k-1}.$$

The formula is true for Q_1 , which has one edge. Assuming $|E(Q_{k-1})| = (k - 1)2^{k-2}$, the recurrence yields

$$|E(Q_k)| = 2(k - 1)2^{k-2} + 2^{k-1} = k2^{k-1}.$$

Thus the result follows by induction.

- (b) View the vertices of Q_k as binary strings of length k , with adjacency when two strings differ in exactly one coordinate. Let x and y be distinct vertices having a common neighbor z . Since both x and y differ from z in one coordinate, x and y must differ in exactly two coordinates. A common neighbor is obtained by changing in x one of those two coordinates, so there are exactly two possible common neighbors. (If x and y do not differ in exactly two coordinates, they have none.) Consequently two distinct vertices of Q_k can never have three common neighbors. In a copy of $K_{2,3}$, however, the two vertices in the part of size two have all three vertices of the other part as common neighbors. Therefore no Q_k contains $K_{2,3}$. □

4. Prove that every edge in the Petersen graph G belongs to exactly four 5-cycles, and use this to show that the Petersen graph has exactly twelve 5-cycles. (Hint: For the first part, extend the edge to a copy of P_4 , and apply Proposition 1.1.38.)

Proof. Use the standard realization of the Petersen graph whose vertices are the 2-subsets of $\{1, 2, 3, 4, 5\}$, with two vertices adjacent when the subsets are disjoint. Permutations of $\{1, 2, 3, 4, 5\}$ induce automorphisms, and any edge can therefore be carried to the edge $\{1, 2\}\{3, 4\}$. It is enough to count the 5-cycles through this edge.

The neighbors of $\{3, 4\}$ other than $\{1, 2\}$ are $\{1, 5\}$ and $\{2, 5\}$, while the neighbors of $\{1, 2\}$ other than $\{3, 4\}$ are $\{3, 5\}$ and $\{4, 5\}$. Choosing one vertex from each of these two pairs determines the middle vertex of a path back from $\{3, 4\}$ to $\{1, 2\}$: it must be the unique 2-subset disjoint from both chosen vertices. The four choices give middle vertices $\{2, 4\}$, $\{2, 3\}$, $\{1, 4\}$, $\{1, 3\}$, respectively. Thus there are exactly four 5-cycles through the fixed edge, and hence through every edge.

Now count incidences (e, C) in which e is an edge lying on a 5-cycle C . The Petersen graph is 3-regular on 10 vertices, so it has 15 edges. Counting first by edges gives $15 \cdot 4 = 60$ incidences. If there are N 5-cycles, counting by cycles gives $5N$ incidences. Therefore $5N = 60$, and the Petersen graph has $N = 12$ 5-cycles. □

5. Use complete graphs and counting arguments (not algebra!) to prove that

- (a) $\binom{n}{2} = \binom{k}{2} + k(n-k) + \binom{n-k}{2}$ for all integers $0 \leq k \leq n$. (Hint: Count the edges of the complete graph K_n in two different ways.)
 (b) If $\sum n_i = n$ then $\sum \binom{n_i}{2} \leq \binom{n}{2}$

Proof. (a) Partition the vertices of K_n into sets A and B of sizes k and $n-k$. There are $\binom{k}{2}$ edges with both endpoints in A , $k(n-k)$ edges with one endpoint in each set, and $\binom{n-k}{2}$ edges with both endpoints in B . These three disjoint classes contain every edge of K_n . Since K_n has $\binom{n}{2}$ edges, the identity follows.

- (b) Partition the n vertices of K_n into sets of sizes $n_1, n_2, \dots$. Within the i th set there are $\binom{n_i}{2}$ edges, so $\sum_i \binom{n_i}{2}$ counts all edges whose endpoints lie in the same part. These form only a subset of the $\binom{n}{2}$ edges of K_n ; the remaining edges join distinct parts. Hence $\sum_i \binom{n_i}{2} \leq \binom{n}{2}$. □